

Java User Group Ingolstadt e.V.

Unleash your Personal Energetic Transition

with IoT devices,
balcony power plants,
Java and Cloud



Disclaimer

The information relating to laws and standards in this presentation has been carefully researched and compiled to the best of my knowledge and belief.

However, I cannot guarantee its legal correctness.

The content is in no way intended to be an advertisement for products or specific companies. It all relates to my specific experiences that I would like to share with you.

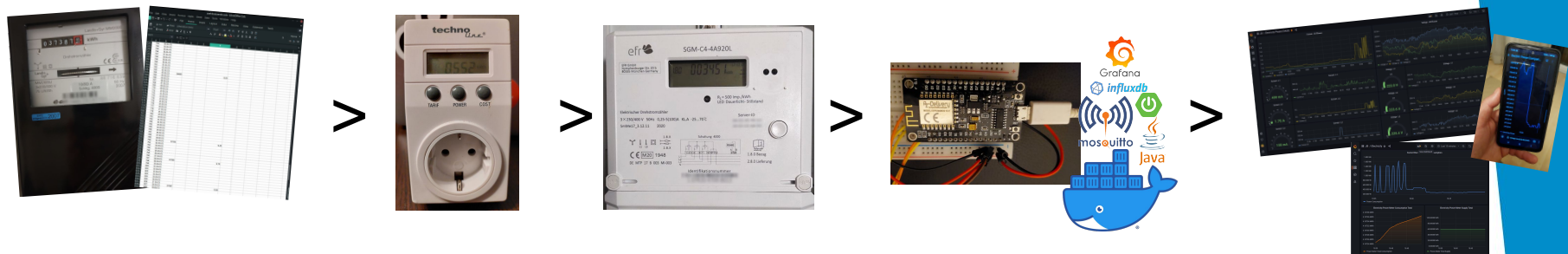
*“Measure what is measurable -
making measurable what is not measurable”*

Galileo Galilei, 1564 - 1641 jul. / 1642 greg.

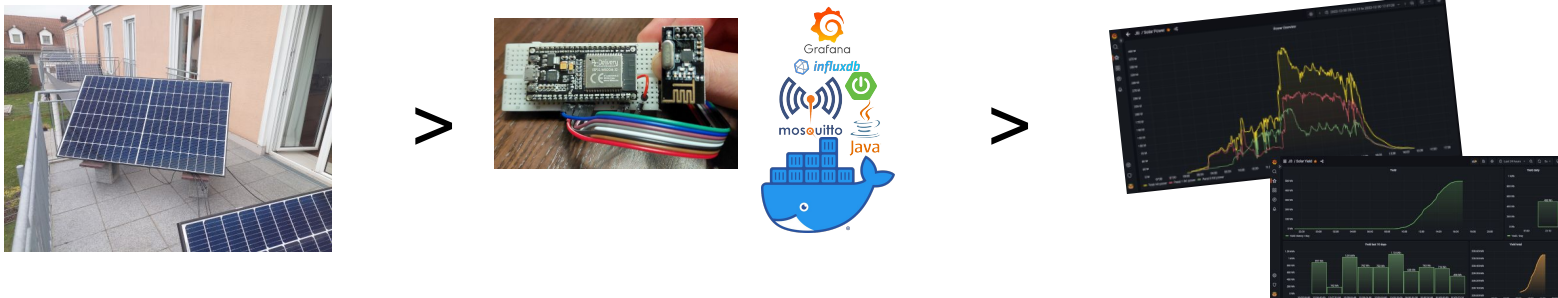
Agenda

Unleash your personal energetic transition, by

1. Making power consumption easy to measure and visualize in real time



2. Generating your own electricity, make it easy to measure and visualize it in real time

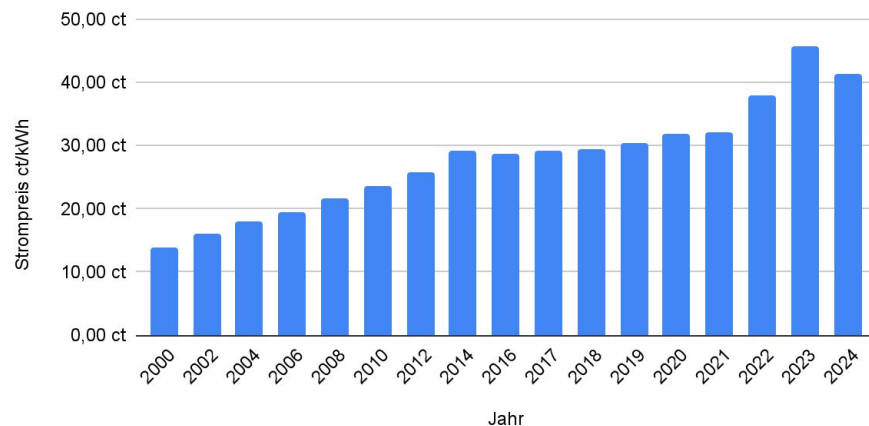


Motivation / Initial Situation

- The will to save energy is probably existing for most people:
 - Climate crisis
 - Electricity prices continue to rise

Strompreis ct/kWh und Jahr

Quelle: BDEW-Strompreisanalyse Dezember 2022



- But: Once all light bulbs have been replaced by LEDs, what to do?

Spoiler: I found a way to save 34% p.a.

- Solution is **saving** and **producing** energy

Ihre **Verbrauchsübersicht** vom 10.10.2023 für den Zeitraum 01.10.2022 bis 30.09.2023. Seite 3

Strom

Vertragsnummer:
Lieferstelle: **85077 Manching**
Vertragsgegenstand: SWI RegioVolt (AGB)
Belieferung außerhalb der Grund- und Ersatzversorgung
Vertragsdauer: 11.11.2023, danach jeweils Verlängerung um 1 Tag(e)
nächstmöglicher Kündigungstermin: 11.11.2023
Kündigungsfrist: 1 Monat(e) zum jeweils nächstmöglichen Kündigungstermin.
Marktllokations-ID: 503
Netzbetreiber*: 9901068000001 - Bayernwerk Netz GmbH
Messstellenbetreiber*: 9906532000008 - Bayernwerk Netz GmbH
*Codenummer gemäß Marktpartnerverzeichnis BDEW bzw. DVGW

Messlokations-ID:	DE001 		
Zählernummer:	1EFR 	Messart: T0FM	
		OBIS-Code: 1-0:1.8.0	
Zählerstand am: 30.09.2022	3.491 kWh		
Zählerstand am: 31.12.2022	4.306 kWh	Ablesekennz.: Ablesung Netzbetreiber	
Differenz:	815 kWh =		815 kWh
Zählerstand am: 01.01.2023	4.306 kWh		
Zählerstand am: 28.02.2023	4.523 kWh	Ablesekennz.: Hochrechnung	
Differenz:	217 kWh =		217 kWh
Zählerstand am: 01.03.2023	4.523 kWh		
Zählerstand am: 30.09.2023	5.153 kWh	Ablesekennz.: Hochrechnung	
Differenz:	630 kWh =		630 kWh

Berechnungsgrundlage für den Zählerstand am 30.09.2023 ist der Zählerstand 5.106 kWh am 11.09.2023.

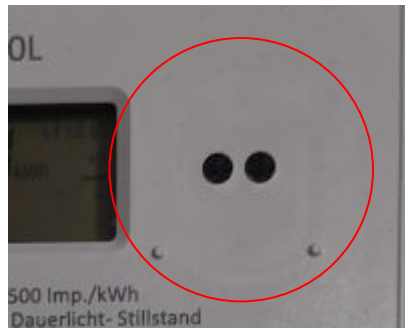
Verbrauch	vergleichbarer Zeitraum Vorjahr 01.10.2021 - 30.09.2022	letzte Abrechnung	aktuelle Abrechnung
Menge HT	2.532 kWh	2.532 kWh	1.662 kWh

Savings:

- ~ 870 kWh of electric energy
- ~ 340 EUR saving *
- ~ 300 kg of CO2 **

each year

Digital Power Meters - “Moderne Messeinrichtung”



Modern power meters have an infrared data interface via which the measurement data can be read out.

To transfer all data, the “Inf” submenu may need to be activated.



A PIN is required for all data that goes beyond the consumption and feed-in meter readings.

This can be obtained from the grid operator.

<https://www.bayernwerk-netz.de/de/energie-service/messstellenbetrieb/pin-prozess.html>

Reading data via the IR interface

Cheap and easy solution to read the data: Hichi reader head...



-TTL-
-Hichi-

ttl ir lesekopf lese-schreib-Kopf EHZ Volkszähler original-Hichi Smartmeter

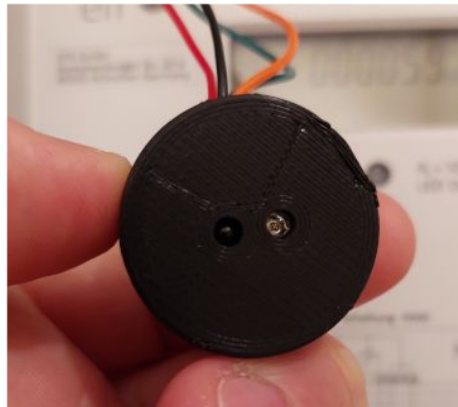
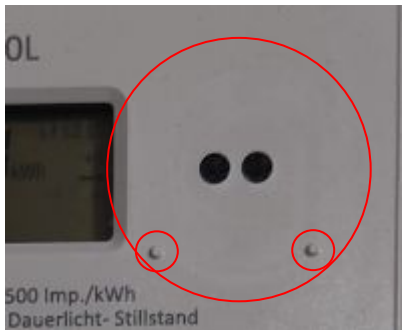
Liest uns sendet SML OBIS DO S0 --Schneller Versand--
Brandneu | Gewerblich

EUR 18,90

Sofort-Kaufen
+EUR 1,55 Versand
Verkäufer zahlt Rückversand

Bis 7% sparen mit Multi-Rabatt

Anzeige



...driven by a ESP8266 Microcontroller

Microcontroller for capturing sensor data

Espressif ESP8266 und ESP32 super suitable for IoT data capturing:

- lots of supported buses
- Wifi capable
- low energy consumption
- MCUs cheap (~4€ / ~9€)

		
MCU	ESP8266 12F	ESP32 NodeMCU WROOM32
Clock frequency	80MHz / 160MHz	80MHz / 240MHz
RAM	Instr 32kB, Data 96kB	512kB
Ext Flash	4MB	4MB
GPIO	17, ADC 1, UART 2	34
Interfaces	SPI, I ² C, I ² S, UART	SPI, I ² C, I ² S, CAN, UART
WiFi protocols	802.11bgn	802.11bgn, 150Mbps max
Bluetooth	-	4.2 Classic, BLE

Tasmota - Open Source Firmware for ESPx

Open Source firmware for ESP8266 / ESP32

- Easy Configuration via Web UI
- Over the Updates possible
- Completely local (!) control
- Supports MQTT, HTTP, serial communication, KNX



<https://github.com/arendst/Tasmota>

19k Stars 4,2k Forks

“Balcony” Power Plants



Energy generation on the balcony

In order to further minimize energy purchases, it has been possible in Germany to operate small solar systems of up to **800W** since 2024.

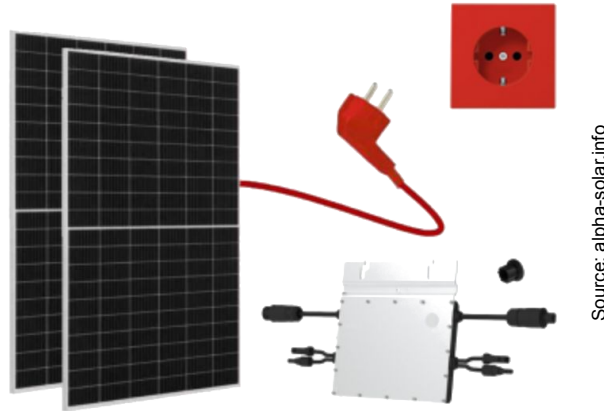
- For personal use only
- No feed-in tariff



From my own experience, a generation capacity of about **670 kWh p.a.** can be achieved under good circumstances

Balcony Power Plant Components

- Power generating equipment, usually consisting of
 - 1-2 (sometimes 4) solar panels (max. 2000W peak total currently)
 - 1 micro inverter (max. 800W feed-in power AC)
 - Ready to plug in with Schuko or power ingress socket
- Feeds the generated energy into the house/power grid via the socket



“Energy Socket”

- The VDE standard speaks of “energy socket”, which enables a contact-free connection
- What is actually meant is the product of a single manufacturer
 - Wieland RST20 Flush-mounted, street price 27€ + shipment
 - Wieland RST20 Surface-mounted, street price 20€ + shipment



Micro inverter

- Converts DC from solar panels into 230V / 50Hz AC
- Requires grid voltage to produce electricity
 - Security feature
 - Required due to grid frequency synchronization
 - Without grid voltage, no solar power is fed into the grid!

Therefore, the market for microinverters is manageable, common:
Envertech, Hoymiles, AE-conversion, APSystems, Huayu

Components and Street Prices

- Solar modules, small quantities street price excl. shipment
 - Canadian Solar, 410Wp ~65€ per piece
 - Trina Solar, 500Wp double glass ~110€ per piece
- Micro inverters, street prices excl. shipment
 - Hoymiles HMS-400, 1 Panel ~80€
 - Hoymiles HMS-800, 2 Panels ~170€
 - Hoymiles HMS-1600, 4 Panels ~230€
- Grid conn. cable Betteri / Wieland or Schuko ~50€
- Mounting ~50 - 100€
- Shipment for solar panels quite expensive (60-80€), better avoid

There are cheaper complete packages, e.g.

800W: 2x Trina Solar 430Wp + Hoymiles HMS-800 for ~350€

Assembly of the balcony power plant

- Assemble the stand / mount
- Install microinverter, connect panel(s) with the MC4 connectors
- Set up, **don't forget to add weights** if not anchored (wind load)!
- Connect to “power socket”



OpenDTU

- A discussion became a software project on GitHub

“Software for ESP32 to talk to Hoymiles Inverters”

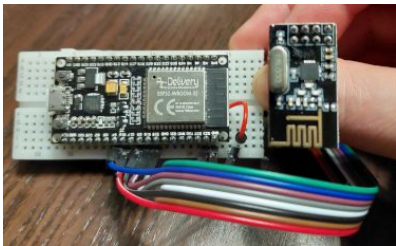
<https://github.com/tbnobody/OpenDTU>

- Free system to manage up to 10 inverters per unit
 - Web UI
 - Rest API
 - Prometheus endpoint
 - MQTT interface with TLS-Support (Let's Encrypt supported)
- Runs on ESP32 Microcontroller
- Requires an additional radio module NRF24L01+ or CMT2300A for connecting with the microinverters

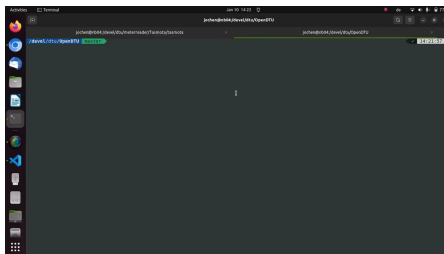
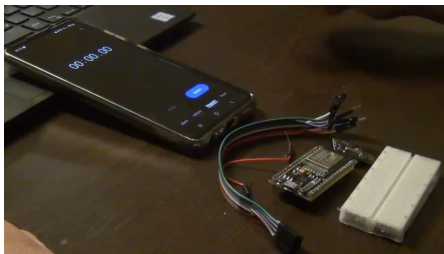
OpenDTU assembly

Complete instructions and circuit diagram on the GitHub page

- ESP32 NodeMCU ~9€
- NRF24L01+ ~6€ - CMT2300A ~ 5€
- Jumper cables male/female, bread board



Indoor model "Raww"



Outdoor model "IKEA lunch box"

OpenDTU Rest API

- All values / settings are also available via Rest API
- Responses here are structured JSON messages

https://www.opendtu.solar/firmware/web_api/

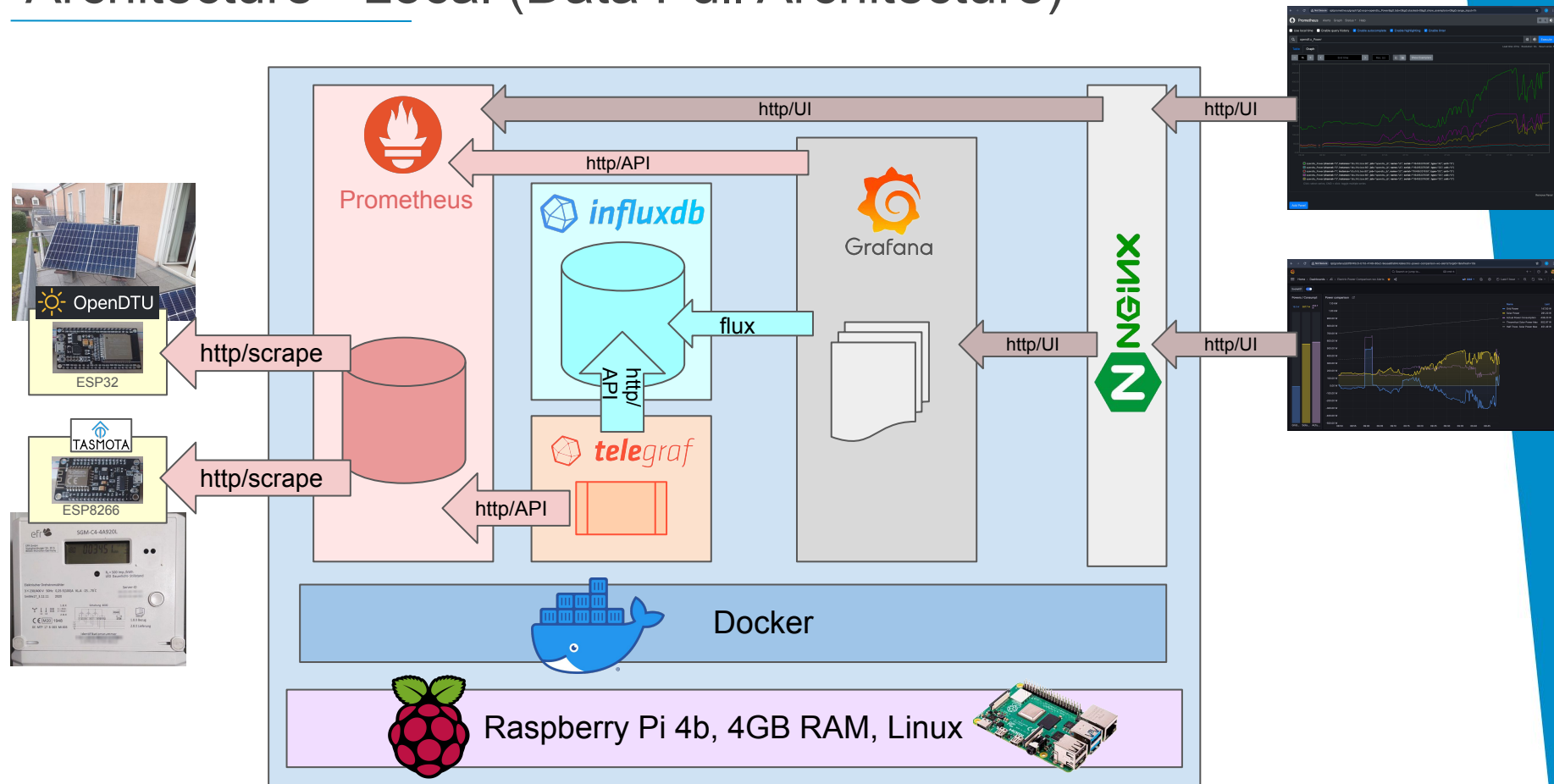
GET/POST	Auth required	URL
Get	yes	/api/config/get
Post	yes	/api/config/delete
Post	yes	/api/config/upload
Get	no	/api/devinfo/status
Get+Post	yes	/api/dtu/config
Get	no	/api/eventlog/status?inv=inverter-serialnumber
Post	yes	/api/firmware/update
Get	yes	/api/inverter/list
Post	yes	/api/inverter/add
Post	yes	/api/inverter/del
Post	yes	/api/inverter/edit
Post	yes	/api/limit/config
Get	no	/api/limit/status
Get	no	/api/livedata/status
Get+Post	yes	/api/mqtt/config
Get	no	/api/mqtt/status
Get+Post	yes	/api/network/config
Get	no	/api/network/status
Get+Post	yes	/api/ntp/config
Get	no	/api/ntp/status
Get+Post	yes	/api/ntp/time
Get	no	/api/power/status
Post	yes	/api/power/config
Get+Post	yes	/api/security/password
Get	no	/api/system/status

- Settings such as feed-in limits can also be set via POST calls

Time for visualization

- Dashboards: Grafana 
Grafana
- Timeseries-DB: InfluxDB 2.x  *influxdb*
- MQTT Broker: Eclipse Mosquitto 
mosquitto
- Containerization: Docker 

Architecture - Local (Data Pull Architecture)



Cloud based (minimal)

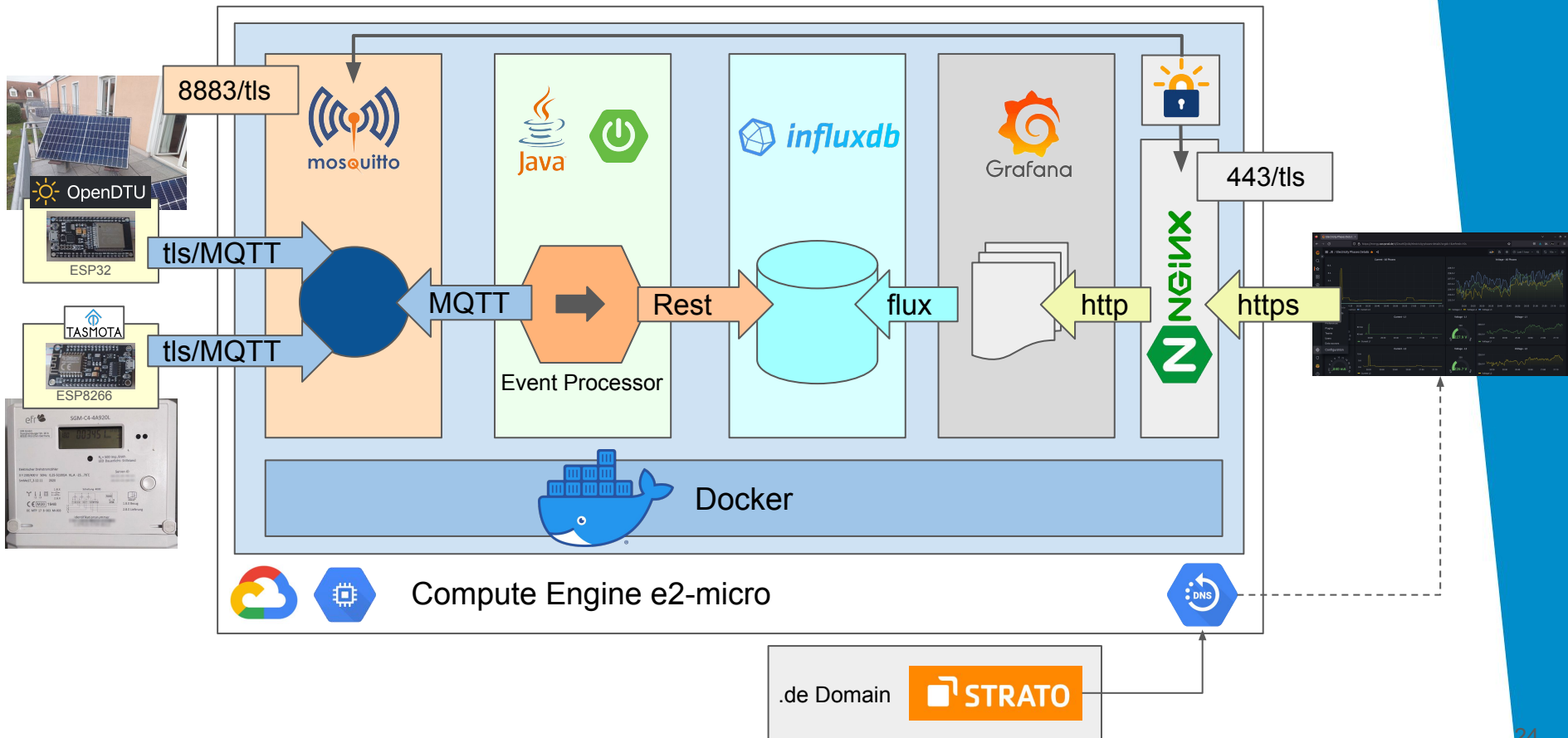
Implemented solution: Google Cloud Free Tier 

<https://cloud.google.com/free/docs/free-cloud-features>

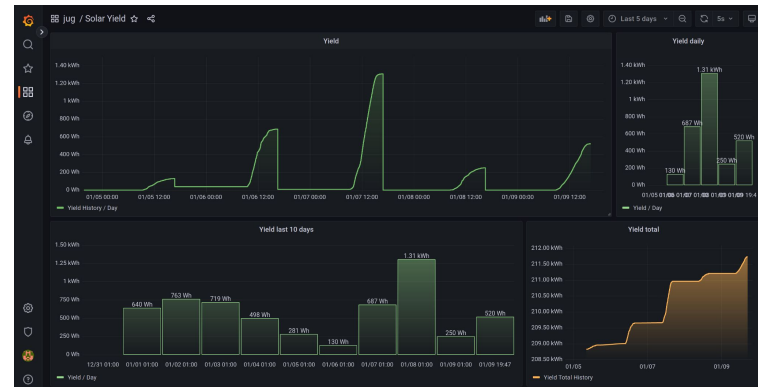
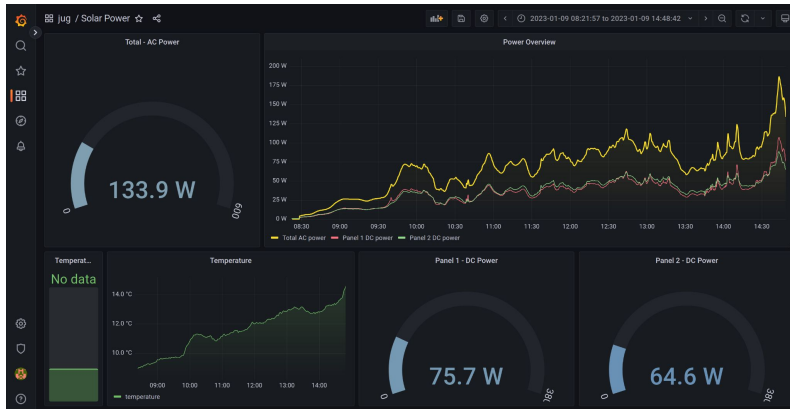
Comprises without cost:

- e2-micro VM in us-east1 (US, South Carolina)
 - 2 vCPUs, 1 GiB RAM
 - 30 GB Persistent Disk
 - 1 GB network egress

Architecture - Cloud (Data Push Architecture)



Data cinema



- All data on inverters and panels
 - All currents and voltages
 - Direct Current (DC)
 - Alternating current (AC)
 - Yields and trends
- visualizable in real time

Conclusion Balcony Power Plants

- Simple and cost-effective entry into climate-friendly electricity generation
- Solar panels are expected to have a lifespan of > 20 years
- Inverters should last 10-15 years
- Amortization
 - Approximately **680kWh** are generated per year
 - About 70%-80% of that energy is consumed normally (depends)
 - Electricity working price currently ~0,40€/kWh
 - 800W balcony power plan purchase price ~**370€**

>> Payback in less than 3 years

- The opportunity is really absolutely unbelievably good right now

Have fun halving your electricity bill!

